

MAE 321 Wing Project Preliminary Design Report

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Abstract

The objective of this project was to design and manufacture a load-bearing wing capable of holding a load of 95.5 pounds applied 23" away from the secured end of the wing. Beyond merely holding the load, the group sought to minimize the weight and deflection of the wing under the applied load. To achieve this objective, the group has begun experimenting with various designs. These designs led to variations in the primary beam, the bulkheads within the wing, the characteristics and surface area of the wing's skin, and the rivets used to hold the skin in place. Beyond merely altering the physical design and dimensions of these parts, the group also examined the different materials that these parts were constructed out of. After many rounds of analysis, it was determined that a 3/4" X 3/4" I-beam constructed out of 7075T6 Aluminum, combined with 9 evenly spaced bulkheads, and 0.04" skin thickness of 6061T6 Aluminum on both sides of the wing would minimize weight and deflection while still handling the applied load. The final cost of the wing was approximately \$267.30, but significantly more material was purchased than was used in the final construction of the wing. The wing weighed 1.88 pounds when finalized and mounted onto the testing apparatus. When tested, the wing held 155 pounds and failed when a 180-pound load was applied. This gave the wing a load-to-weight ratio of 82.4. The wing performed well, holding the desired load with a reasonable safety factor. Further steps to improve the design would focus on ensuring that the rivet number and placement is more ideal to decrease the deflection. Another area for improvement might be in altering the central beam to reduce weight.

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Introduction

This project's objective was to design and manufacture a lightweight wing that could support at least a 95.5-pound load applied at the tip of the wing. The wing was 22 inches long, with a 1 ½ inch protrusion extending from the end. This brought the total length of the design to 23 ½ inches. In addition to this, a horizontal 3/16" hole was drilled 23 inches from the securing end. The beam itself extended two inches from the large end to be securely fixed under load. This two-inch extension was a ¾" square segment, with a ¼" hole drilled vertically 1" away from the first bulkhead. The area of the wing had to fall between 98 and 102 square inches per side. The design of the airfoil was a symmetric airfoil provided to the group. Notably, the shape of the airfoil could not be changed, only the scaling factor between the beginning and the end of the bulkheads to which the wing is attached.

While designing the wing, 4 primary components needed to be considered. These components were the main beam, the bulkheads, the skin, and the riveting regime. Optimizing each of these components was crucial in determining the ideal design that could support the necessary weight while also attempting to minimize the deflection of the wing under load and reduce the weight of the wing as much as possible. An I-beam that extended along the entirety of the wing was crucial to support the bulkheads and skin. The bulkheads served to connect the skin that covered the airfoil to the beam, and the rivets ensured that the connection between these components was firm. The ability of these components to work together was crucial in creating a good wing design.

To ensure that the wing could withstand the applied load, the group took advantage of the finite element analysis capabilities built into the CREO software to experiment with various designs for the main beam, the bulkheads, skin characteristics, and rivet placement. The designs were optimized over many iterations as the group sought to maximize the potential benefits from each component. To that end, each potential component was examined with static analyses under a standard load of 100 pounds. To ensure that the group emerged having selected the optimal layout for the wing construction, variables were altered one at a time. After identifying the most promising designs that emerged from radical alterations to the designs, sensitivity studies were used to further optimize results. Each part was optimized independently before being integrated into a single consolidated part for final testing and confirmation of the design. The details on the final selections for these components are highlighted in the following sections.

As the group manufactured the different components that made up the wing, the group used many different tools within the shop. This included the manual milling machines, the computer numerical control (CNC) mill, hand drills, rivet gun, waterjet cutter, 3D printers, and

the band saw. The manual mill was used to prep the beam with mounting holes. The beam was then mounted in the CNC mill, where it was taken down to the correct dimensions for use in the wing. The waterjet cutter was used to cut the two bulkheads at the ends of the wing. The 3d printer was used to print the middle bulkheads and the leading edge. The band saw was used to cut the skin for the wing, while the hand drills and rivet guns were used to mount the different components together. Notably, whenever issues arose during the manufacturing process, the design was refined and altered slightly to ensure that the final product was still capable of holding the necessary load. This included the specific placement of bulkheads, as well as rivet placement and length, which was optimized to provide the maximum attachment while not compromising the integrity of the other components.

Finally, the wing was tested and compared to the other group's results. The wing succeeded in holding loads of 65, 100, 125, 135, and 155 pounds before destructively failing at 180 pounds. This met project specifications, and the group successfully completed the necessary task of carrying a load on the wing. Notably, the deflection of the wing was quite large.

Design Theoretical Analysis

Our group was split into 5 subteams, each of which had specific tasks to complete. We chose the groups to be the main 4 parts of the wing project, along with the Team Leads. The Team Leads helped to keep deadlines for the group and assisted any other group members with their specialization. They also helped communicate details and design ideas between the other groups to ensure everyone was kept in the loop for large, overarching design changes.

Role	Member 1	Member 2
Team Leads	Jack Brynes	Lucas Denault
Bulk Heads	Raj Patel	
Rivets	Esteban Nunez	Sawooly Li
Skin	Connor Brown	Ani Kozak
Beam	Tyler Harris	

Table 1: Tasks and Role Distributions

All design work, modeling, and simulation were done through Creo, which enabled us to analyze stresses, deflections, and component behavior as we ran through different design options.

Our main goal was to design a structure that met the project's load requirements while ensuring it would still be within the 98-102 sq in surface area. Although there were no strict cost limitations, we still aimed to make material choices using McMaster-Carr that made sense for the

project. Materials were selected mainly based on their mechanical properties and by optimizing load displacement as determined by Creo simulations. For example, we realized early on that the beam required a material that was rigid and firm yet still lightweight, so we chose AL7075. The skin material needed to be thin and easy to rivet, so we chose the AL6061.

To optimize the structure, we varied dimensions and features like the flange, bulkhead spacing, and skin thickness (and material). Each component was tested individually in Creo by our subteams, and we iterated the tests a lot to ensure that we had optimum force displacement and make sure that there were no breaking points on any of our components. To know the breaking points, Glenn suggested some ksi we should try to be under for each part. For example, he told us that the rivets would fail around 20 ksi, so when doing our analysis, that was our target number to be under. For all of the different subteams, we tested different geometries and layouts, taking away material that wasn't greatly contributing to the structural properties.

Looking to the future, the group considered the manufacturing side of the project while finalizing the wing's design. Some designs were slightly altered so that they could be manufactured more easily with the CNC and other equipment in the shop.

Materials Selection and Bills

Wing Skin:

Multipurpose 6061T6 Aluminum Sheet: 0.040" Thick, 24" x 24" - \$40.98

Part Number: 89015K42 [Link](#)

Bulkheads:

Bulkheads on both Ends (2):

Multipurpose 6061T6 Aluminum Bar: 6" x 6" x 3/4" – \$34.63 x 2

Part Number: 8975K235 [Link](#)

Middle Bulkheads (7):

PLA Plus from a 3D printer located in the MAE shop.

Rivets:

Connect Skin, Bulkhead, Beam:

Aluminum Blind Rivets with Aluminum Mandrel, Domed Heads, 1/8" Diameter
For 0.251"-0.312" Material Thickness 250 ct. – \$12.56 [Link](#)

Aluminum Blind Rivets with Aluminum Mandrel, Domed Heads, 1/8" Diameter
For 0.313"-0.375" Material Thickness 250 ct. – \$16.32 [Link](#)

Connect Skin to Bulkheads:

Aluminum Blind Rivets with Aluminum Mandrel, Domed Heads, 1/8" Diameter

For 0.126"-0.187" Material Thickness 250 ct. – \$10.06 [Link](#)

Beam:

High-Strength Aluminum 7075 Bar: 1" x 1" x 36"

Part Number: 9055K47 – \$118.12 [Link](#)

Bill:

\$40.98 + \$34.63(2) + \$12.56 + \$16.32 + \$10.06 + \$118.12
= \$267.30

Procedure

In this section, we will detail the engineering design process and the thought behind our final models. The models will be separated into subgroups to explain individually why each group thought how they did and reached the final conclusion. As a side note, all figures from CREO will be referring to those in Appendix A, while all drawings will be located in Appendix B.

Section 1: Beam

Many types of beams could potentially be used in this project. The group briefly considered the pros and cons of several different types of beams, including circular beams, C beams, and I beams. When analyzing each of these types of beams, the group took into consideration several key factors. The first consideration was what kinds of loads the wing is expected to withstand. Because the load is being applied approximately directly downwards, it is expected that the beam will need to be capable of supporting large bending moments and shear forces. Due to the vertical application of the load, it is anticipated that there should be minimal torsional forces. Because of these requirements, an I-beam design was determined to be optimal.

In designing the I-beam, there were several key considerations that needed to be addressed before reaching a final design. These factors included: Dimensions of the beam at the beginning and end, thickness of the web, thickness of the flange, material, and weight reduction. By optimizing each of these factors, the design of the beam should be very successful when the force is applied to it.

The first consideration when designing the beam was determining the material that it should be constructed out of. This was a simple task, and it was accomplished by creating a simple model where the different materials could be tested for displacement, weight, and stress. In this model, numerous different materials were tested to examine how they would deal with the fully applied load. From this analysis, it was found that the most ideal material would be aluminum 7075 T6. It was chosen for being relatively lightweight, possessing the same toughness as 6061T6, while having greater strength. Despite weighing more than 6061T6, the

7075T6 has a significantly higher yield strength, enabling it to carry more of the load should the rest of the wing fail. As seen in Figure 1A, the maximum stress the initial beam could withstand was 74.3054 ksi, and in the mass optimized beam, Figure 1B, we found the maximum stress to be 67.5467 ksi. And despite weighing more, selecting the more resilient 7075T6 should make it possible to remove more from the beam itself, further reducing the weight of the beam.

When considering the scaling of the beam, there are several key considerations. On the one hand, having a larger beam means that the stress on the beam will be minimized, though the weight of the beam will increase. On the other hand, if the beam is too narrow, it will be unable to support the load and cause the wing to fail. To combat this, several different dimensions were analyzed over the course of designing the beam. From these analyses, it was clear that the best solution was to leave the beam at $3/4" \times 3/4"$ at the mounting point and scale it down to $3/8"$ by $3/8"$ at the end of the wing. Choosing these dimensions made it possible to minimize the mass of the beam while ensuring the maximum stress was reasonable.

Now, considering the web and flange thickness, it was initially decided to keep the web and flange thicknesses equal. Analysis was then used to determine what the optimal thickness ratios would be between the two in order to prevent the beam from being too heavy, while also ensuring that the beam would be capable of holding the necessary weight from the load. After determining the correct thickness, a final analysis was run. Upon running the analysis it was clear that the center of the beam did not experience any significant stresses, and the higher stresses were felt by the top and bottom surfaces of the beam. As such, it was determined that removing material from the central web of the beam would be an ideal way to reduce the weight of the beam without weakening the wing significantly. To that end, the web thickness was decreased incrementally. There appeared to be no real changes in the beam regardless of how thin the web got, however, there were concerns that if the web were too thin, the beam could fail along the axis. This could occur due to the forces not being perfectly centered, or other minor issues that might cause a catastrophic failure if the web were not strong enough. Beyond merely shrinking the web thickness, there were also holes put into the beam along the vertical axis. Once again, these holes were analyzed to ensure that the beam would still be capable of carrying the weight. When creating the holes on the beam, we used an equal constraint to make the distances between consecutive holes equal, as well as the distances between the top of each respective hole and the bottom of the flange equal. After running repeated analyses, we also found that when we decreased the distance between the top of each hole and the bottom of the flange - which made the hole diameters larger - this did not make the stress felt by that region change much, so we felt comfortable making that distance between the flange and the top of the holes relatively small in order to reduce weight. We also ran analyses to analyze the stress between holes when the distance between holes changed, and we found 0.1" to be the optimal distance between holes in order to optimize weight reduction and beam strength. Also, we changed the sharp corners of the I-beam cross sections into fillets since fillets have been shown to experience less stress than sharp corners, so this change improved the stability of the beam. After completing these weight

removal measures, it was determined that the beam was still strong enough to meet the demands for the wing, while being lighter in order to save weight.

Section 2: Rivets

When analyzing the rivets, CREO presents a multitude of challenges that make modeling and analysis difficult. Because of this, many assumptions were made to analyze the most representative model of our wing within the constraints of CREO's simulations. The fastener feature could not be applied to our actual wing assembly for a variety of reasons. Primarily, the fasteners within CREO cannot interface surfaces that are not flat against each other. Therefore, the curved nature of our wing design would not work for our rivet analysis. Therefore, a duplicate wing model was created to replicate our wing design as closely as possible while still enabling the fastener analysis to be done. Within this duplicate model, a variety of assumptions were made, which are discussed below:

1. To get around the curve & fastener interference issue a duplicate model of the wing was created. The duplicate model (Figure 2A) is essentially a low poly model of the real wing. It consists of four primary parts – a bulkhead part consisting of the nine separate bulkheads (Figure 2B); the flat beam model (Figure 2C), see assumption (3) for a discussion on why it is flat; a part consisting of the curved part of the wing (Figure 2D); and a flat top wing surface (Figure 2E). Of course, this wing would not fly but it allows us to insert fasteners between the bulkheads and surface of the wing since all part surfaces interacting with the fastener are perfectly parallel – Assumption (1) the duplicate wing model is entirely flat.
2. This wing duplicate is one-sided, that is it does not have a bottom surface to it. This is due to another limitation of the CREO fastener feature, in which fasteners cannot intersect three parts – Assumption (2) no bottom skin.
3. Similar to assumption (1) there is yet another issue in which our original beam design is on a slight curve throughout the entirety of the wing. This is a feature of the swept blend design which means the beam cross section at one end is $.75 \times .75$, and then narrows to $.375 \times .375$. This means that the beam's top and bottom surfaces are not flat; this presents an issue in which any extrusion through this part would result in an oval shaped hole, which CREO's fasteners can't interact with. So, to get around this the beam within the wing model is not designed with a slant, that means it is $.75 \times .75$ throughout the length of the wing. This assumption may initially seem as a dangerous one to make, as it would drastically increase the strength of the model towards the outer wing of the tip, as the wing is two times its actual cross sectional size – but because this model is being used only for rivet analysis it is not an issue. That is due to the fact that the first couple

bulkheads are the focus of the analysis, and at these points the beam duplicate is a very similar size to the actual beam – Assumption (3) beam cross-section is .75 x .75 along the entire length.

4. In order to fasten the beam to the bulkheads, without introducing further issues due to using the fastener feature and introducing another dimension of unnecessary complexity to this analysis, the bond feature of CREO was utilized. This means that there is no stress showcased at the interface between the bulkheads and beam. This assumption is okay to make since the stresses experienced by these bulkheads will be much less than those experienced by those on the wings surfaces – Assumption (4) the beam and bulkheads are bonded rather than fastened.

The final duplicate model of the wing, as seen in Figure 2E consists of three separate parts: the top wing surface, the bulkheads, and the curves of the wings. This model was then utilized to conduct a multitude of analysis, in which the location and amount of rivets was altered. The analysis was run with 90 lbf on the far tip of the beam, with the opposite side of the beam (closest to bulkhead 1) being constrained, as well as bulkhead 1 itself. The final analysis, as seen in Figure 2F, consists of the wing model with a total of 34 rivets on the top surface. Due to the low levels of stresses seen in Figure 2F, specifically along the top surface of the wing along bulkheads 4 through 8, those bulkheads will have only two rivets instead of the 3 seen in the design. The analysis shows that the rivets attached to bulkhead 1 will experience the greatest stresses, reaching nearly 25 ksi – rivets will fail at 20 ksi, but because the model does not take into account a bottom skin, the peak stress experienced by these rivets should drop to under 20 ksi.

Section 3: Skin

The third major component of the wing design process involved evaluating and optimizing the skin thickness and material. Initially, a uniform skin thickness of 0.03” was selected for both the top and bottom surfaces of the wing. This thickness was tested using both 6061 and 7075 aluminum alloys to evaluate the load distribution and the flexing of the wing as a whole. Our goal was to have the total deflection less than 1 inch, and the entire distribution be below 35 ksi. After those were met, we attempted to reduce the weight by lowering the thickness.

After doing some analysis, the group saw that deflection values were consistently exceeding 1 inch across all types of materials with a thickness of .03” (even making the top and bottom surfaces different materials) as can be seen in Figure 3A and 3B. In Figure 3A, there is an analysis of this excessive deflection when the material was AL6061 on the top, and AL7075 on the bottom. Due to this, we decided to have the thickness at .04” all around to ensure that the deflection is less than 1 foot, regardless of what we decide upon for the skin material.

With the thickness fixed at 0.04", a final material comparison between AL6061 and AL7075 aluminum was conducted. Here is where we started experimenting with making the bottom skin AL7075 and the top AL6061, and vice versa. The experiments were first fueled by knowing that the top skin would have to bend more than the bottom, so we wanted a rigid bottom to fight the deflection, with a top surface that would bend with the change a little more than its hardened counterpart. So, we initially tried the bottom skin with AL7075 and the top with AL6061. After doing this analysis, and comparing to the base AL6061 all around, we saw that the stress and deflection was actually better. As shown in Figure 3D, the deflection is 0.89565" (in Figure 3F the deflection is 0.90340"). As a point against AL7075, the stress analysis in Figure 3C showed that with such a hard material, very high stresses arose in the first and last bulkheads. While the 6061 top and bottom skin design in Figure 3E did not show nearly the same stress levels in the bulkheads, it did deflect a bit more, as seen in Figure 3F, but only by about 0.01 inches. So in the end, the 0.04" 6061 top and bottom skin design served as a less stressful, almost equally deflective, and lighter solution as opposed to using AL7075 for either the top or bottom skin.

While AL7075 offered improved stiffness, AL6061 was ultimately selected for both the top and bottom skin due to its greater ductility, which gave us the give that the skin required for the bend that the wing would experience. Thus, the final design utilized AL6061 aluminum at a uniform 0.04" thickness across all external surfaces. As in Figures 3E and 3F (see Appendix A), the final design saw maximum stresses appear to be around 25 ksi, even at the singularities, with 0.9 inch deflection.

Section 4: Bulkheads

When designing the bulkheads, the team was reliant on other subteams—such as those working on the skin and the beam—before we were able to start the main design. However, we ensured that the beam team would keep the cross section constant, and ignored specifications to the skin until the final product was decided upon. Because of these factors, it was decided to first determine the scaling factor between the first and last bulkheads. Completing this crucial step would determine the total surface area of the wing's skin and denote where the placements for the rivets would be. By using a generic model of the wing and isolating changes to just the bulkheads, it would be possible to determine the most ideal characteristics of the bulkhead subsystem and then integrate this optimized subsystem when developing the rest of the wing elements.

To determine the ideal scaling factor between the two bulkheads, an iterative process was employed. The process began with a wide range of acceptable parameters that still resulted in the total wing surface area falling between the required 98 and 102 square inches. Using numerous static analyses, 100 lb of force was applied to the end of the model wing for each of the different scalings. Doing this essentially allowed a sensitivity study to be conducted, where the optimal scaling factor between the first and last bulkheads was determined. The results from the notable

scaling factors can be found in Table 4.1 in Appendix A. One issue that came up was that singularities did appear in some instances when testing the scaling. Because of this issue, the values were treated as being relative to one another in terms of magnitude, and the deflection was also considered as another potential indicator of what the ideal scaling would be. However, when considering the deflection, wings with a larger skin surface area had an advantage over a smaller surface area. However, because the difference in surface area was relatively small, and it would be possible to have thicker skin on the wing, it was decided that this effect would not alter the final performance of the design. After running many simulations and comparing the results, it was decided that the bulkheads would be scaled by 5.4 and 3.2 on the inside and end, respectively. Choosing these scaling factors resulted in a wing with a surface area of 98.252 inches on each side, falling within the required parameters. Choosing this scaling factor meant that there was significantly less stress and a minimal amount of deflection, although it was not the absolute minimum. The success of the distribution of stress, as well as the comparatively good deflection, can be seen in the simulation results of Figure 4B.

After determining the correct scaling factor of the bulkheads, it was then important to consider the spacing and number of bulkheads within the wing. While the first and last bulkheads have set positions, the intermediate bulkheads have no restrictions, leaving the number and placement to be determined. In order to determine the placement of the bulkheads, numerous analyses were performed. In some cases, the bulkheads were grouped towards the ends, spaced irregularly, and spaced evenly throughout the wing. One example of how bad spacing regime led to poor wing stress distribution can be seen in Figure 4A. After trying these many different layouts, it was decided to proceed with the bulkheads being evenly spaced throughout the wing. By having the bulkheads evenly spaced, it was possible to spread the stress evenly across the wing. Spreading the stress out over the skin will ensure that the wing can withstand more force without yielding.

Now, considering the number of bulkheads present within the wing, it was determined that between 6 and 8 bulkheads was ideal. After further consideration and analysis, it was decided to continue with 7 bulkheads. The reason for choosing to continue with more bulkheads than strictly necessary is due to the fact that the group will not be using the same bulkheads that are being used in the simulation. The changes to the bulkheads will be explained in the next paragraph, but due to this characteristic, the group has decided to utilize an extra bulkhead in order to decrease the stress on each individual bulkhead.

Finally, the group decided to lessen the weight of the wing by altering the design and material choice for the internal bulkheads. In the initial design of the wing, the group had pursued solid aluminum bulkheads that featured only a cutout for the I-beam to pass through. While this has significant strength, it also has significant weight. Upon discussions with Glenn and through alterations of the models, the group determined that the impact of the bulkheads themselves was relatively minimal. To this end, the group decided to look into ways to maximize the impact of the bulkheads while minimizing the weight. From the setup of the force application, the first bulkhead will be extremely important as it will hold the skin in place on the

wing. Similarly, the last bulkhead will also be important, as it again holds the edge of the skin and will be providing the necessary mount point for the compressive force on the bottom of the wing. However, the middle bulkheads will not be as stressed and are therefore a good place to cut weight from. In order to cut adequate weight, the middle bulkheads will be 3D printed and made out of PLA instead of aluminum 6061. Aluminum 6061 is significantly more dense than printed PLA, leading to a large reduction in weight right off the bat. Beyond just reducing the weight by changing the material used to make the bulkheads, some of the material will be removed from the bulkheads. Removing the material from the middle of the bulkheads will not significantly alter the performance of the wing, and will further reduce the weight of the wing. By implementing these changes, the design will remain strong, and become more lightweight. The final design can therefore be seen in Figure 4B.

Further Analysis

For our further analysis, we wanted to look at the fatigue analysis of the beam and the wing, as well as do a modal analysis to see at what natural frequencies the wing would shake at. The fatigue analysis was a little questionable for the beam. We saw that the bottom side of the beam would hit its max fatigue around 2.5 Log Lives, which is a few orders of magnitude less than that of the top (around 5), as seen in Figures 5A and 5B. This does give us some weariness as to the bottom side of our beam. However, when we do the same analysis for the entire wing, we see that the addition of the skin cleans up the fatigue analysis, giving the entire body significantly less chance of failure early on. We see that the middle part of the wing seems to fail around 6.5 Log Life, in Figure 5C, which is much better.

Upon doing the modal analysis, we see that the first modal frequency for the beam is 49 Hz. This means that if you had a piston engine that operated at 49 or a harmonic of that, it would start shaking. This is a simple test to show a practical side of the wing, and can connect to actually putting in the practical section. Further analysis showed that the beam has modes at: 42.8 Hz, 67.6 Hz, 188.2 Hz, and 285.2 Hz, whereas the total wing has modes at: 59.0 Hz, 109.7 Hz, 123.6 Hz, 214.3 Hz

Results and Discussion

In this section, we will review the manufacturing processes and performance of each of the four components of our wing. We will start with a brief overview of the overall performance, followed by explanations of the manufacturing process explaining the details of the procedure. Next, we will evaluate the individual performance of each component and how it contributed to the overall work of the wing. Finally, the conclusion will reflect on our overall strategy, analyzing what worked well and what we would approach differently in retrospect.

Final Results

For the final test, we were happy to see that the wing could carry over 97 pounds, with the final weight of the wing coming out to 1.88 lbs. Below is a table of the weight, and total deflection that the wing experienced, as well as any comments about the damage caused.

Weight (lbs)	Deflection (in)	Comments
65	3	N/A
100	4.625	Bottom skin buckling
125	6	2 rivets popped off
135	6.625	Beam began to twist
155	7	Beam twisting, high buckling
180	N/A	Wing broke

Table 2: Wing Performance

Initially, there was a lot of deflection from our wing, but the progression of events made sense. Slowly, the rivets fell off, and the beam started twisting. At the end, many rivets popped off from the top skin of the wing, the leading edge flew off, most of the middle bulkheads had cracked down the middle, and the beam broke between the first and second weight reduction holes. The damage was as expected, especially with the location of the beam breakage. We will go into further detail in individual sections to assess the components, but overall, the deflection was more than anticipated, and the distribution of the load was as expected.

Section 1: Beam

We had some initial trouble with the manufacturing of the beam. To mount the beam, a series of holes needed to be drilled in a straight line to fit into the mount Glenn created. In our first attempt to drill the holes, we had issues with keeping the line perfectly straight, resulting in issues mounting the beam and removing excess material that could potentially harm the design. Glenn made a valiant effort to salvage the beam and was largely successful, as the beam was mounted and the CNC was able to run its program. However, in running the program, a portion of the beam became too thin, which could potentially have compromised the design of the wing and led the group to fail the load test. Because of this concern, it was decided that the group should try again and begin with a new piece of stock. This time, the holes were drilled accurately, the beam was easily placed into the mount, and the CNC mill perfectly followed its

instructions. There were no major issues with machining the beam this time around, resulting in the manufactured beam matching the simulated CREO beam.

One of the other issues with the beam was related to the bulkheads. While installing the bulkheads, the placement of each was determined by simply sliding the bulkhead along the beam until it became fixed in place. All bulkheads fit snugly in their desired position; the only deviation was with the first bulkhead, at the widest section of the beam. This bulkhead did not settle precisely where we needed it to. This issue was noticed by discovering that the leading edge seemed to be too short, and upon inspection, it was confirmed that the first bulkhead was sliding back a little too far. While the group is not entirely certain why the first bulkhead slid too far back along the beam, there are two primary theories. The first theory is that because the waterjet cutter can only cut a directly vertical line, the side that was cut by the waterjet was the wider side, which enabled the entire structure to slide too far along the beam. The second theory, and the more likely problem, stemmed from the fact that there was an issue when operating the waterjet. While trying to cut out the aluminum bulkhead using the waterjet, there was an issue where the internal cube stopped the head of the waterjet from moving. As a result, the cut through the aluminum was not perfectly centered once the issue was relieved. This lack of centering altered the shape of the bulkhead and may have allowed the bulkhead to move too far up the beam.

Despite the issues faced in the manufacturing process, the beam performed exactly as expected. The truss design helped reduce weight without affecting the beam's load capacity. It broke at 180 lbs, right around the first hole down from the fixed edge (Figure 6C), which is where we predicted it would fail using the CREO simulations. This area took on the most stress, so it makes sense that it gave out there once the load got high enough. This proves that the CREO simulations we ran were accurate, beneficial, and allowed us to improve our design.

Section 2: Rivets

The manufacturing process for the rivets was not difficult, but time-consuming. While there were many places we wanted to rivet, and on the day the group went to rivet the skin in, one of the drills was dead, so we had to share with another group. We riveted using parts that had different grip ranges, using smaller rivets for the smallest bulkheads, and used about seven larger rivets for the largest two bulkheads. This was a very long process, and we ultimately decided to forego rivets on the bottom skin closest to the leading edge, which was a mistake. This is something that should be corrected for if we were to iterate into a new version of the wing, with more rivets being placed closer to the leading edge of the wing to avoid buckling.

While testing the wing, not putting those last few rivets in made the deflection significantly worse. Around the 100 lb mark, the bottom skin started coming off the bulkheads in the same place where we had debated on riveting. This was disappointing because we believe this contributed to our larger deflection at higher weights. Also, early on, two rivets at the bottom popped off, and the next ones to depart from the wing were at the final weight of 180 lb. We

believe that this can be explained by just bad riveting jobs. If the rivets were not all the way in, or if the connection was bad in general, they may be predisposed to failing quicker than the other secure rivets that failed at the same time as the rest of the wing. Notably, each individual rivet can only withstand a certain amount of force, so adding more rivets to the skin and bulkhead system would have potentially improved the max load and deflection of the wing.

Section 3: Skin

The machining for the skin was straightforward; we just bought the correct thickness and cut the shape of the wing out twice with the band saw. We made sure that the bulkheads were moved as far down on the skin as possible before making our cut lines. To ensure the skin on the trailing edge was secure, we simply taped the 2 edges together. This allowed for a clean finish on the back side and allowed the trailing edges to be flush between the top and bottom sides. Enabling the skins to be continuous between the top and bottom, meaning they would flex in unison, which could improve the strength of the wing.

The skin held up well during all the trials. Around the 100 lb test, the skin had some bending and flexion, especially near the leading edge. This was likely because we hadn't fully riveted the bottom side of the skin in that area. Despite this, the force was still fairly well distributed across the surface during the higher load trials. The deflection we saw suggests that the skin was effectively transferring load to the internal structure, but without that final set of rivets, the unsupported section near the leading edge became a weak point. This caused localized bending and could have led to uneven stress distribution under heavier loads. There was very clearly buckling behavior that acted along the leading edge of the wing, which became more pronounced under heavier applied loads. If the edges had been riveted more rigorously and thoroughly, the wing might have experienced less deflection when under load.

Section 4: Bulkheads

The machining for the bulkheads had two different approaches. For the middle 7 bulk heads, we 3D printed them on one plate. This was easy to arrange and splice the plate because of the symmetric nature of the bulkheads and the relative size. The first and last bulkheads were made out of AL6061 and were cut out using the waterjet. The only difference was that the middle bulk had several weight-reduction holes inherent in the design. However, the end bulkheads lacked these holes because it was 6061, and we could not easily decrease the weight without sacrificing structural integrity. In the case of the first bulkhead, the waterjet was used to hollow out some small holes that could be tapped to make it possible to mount the wing to the testing apparatus.

Another part related to the bulkhead was the leading edge. The leading edge was designed by first cutting out the shape of the leading edge from the bulkheads, then copying that same geometry with another swept blend. Then, we offset the outer distance to give it a little

more room and stability on that outer surface. This results in one large part, spanning the entire length of the wing. This was a little complicated because we don't have a big enough 3D printer to simply print one large piece, so we split the leading edge into 3 subspecies. Upon riveting the leading edge into the existing bulkheads, we observed that the leading edge appeared to be too small. After further inspection, we realized that the first bulkhead's square cross-section was too large. This allowed it to be further back on the beam and made the leading edge appear off. This oversight was due to the first bulkhead not sitting precisely where it needed to, so adjusting the bulkhead back to its correct position alleviated the issue with the leading edge. Setting the bulkhead to the correct position (about $\frac{1}{8}$ " forward from its resting position) and then riveting it into the beam there. Then, the leading edge was snug on all the bulkheads and fit quite nicely.

The rest of the manufacturing included riveting the bulkheads into the beam in their desired locations and riveting the leading edge to the bulkheads. During this process, there was some cracking within a few of the plastic middle ones, which was concerning because they had sacrificed some integrity just to ensure that they would not move. We cracked 4 of the 7 bulkheads during the riveting process. 3 while riveting the leading edge, 1 while riveting them into the beam. Some of this was due to oversight on the power of the hand drill, and the weakness of the PLA of the bulkheads and the leading edge. Following a major reduction in the speed of the hand drill when drilling through the PLA, the plastic did not crack, and the integrity of the parts was preserved.

During the performance of the wing, the bulkheads worked as expected. The plastic ones all broke down in the middle, though the rivets attaching them to the beam still held. This shows that at this critical stress, the bulkheads strained too much, and they ripped at the intersection of the rivet. During our final test, the bottom part of the leading edge detached and flew off. The failure of the leading edge showed us the importance of both accurate bulkhead placement and stronger attachment methods for high-load scenarios.

Conclusion

To successfully design and manufacture a wing capable of holding a 95.5-pound load, it was necessary to optimize several different components. These components included a main spar to support the majority of the load, bulkheads to keep the shape of the wing fixed and to help distribute the load across the beam, skin that took on the majority of the load, and rivet placements and counts that ensured firm connections between all of the components. While iterating through various solutions and styles of each component, it was determined that a singular I-beam supported by 9 bulkheads was the ideal setup to carry the load while minimizing the weight of the wing itself.

While working on the design of the beam, many different variables were altered to nail down the correct design. These included the dimensions of both ends of the beam, the thickness of the flange and web, and the placement of weight-reducing features such as holes in the beam.

Optimizing all of these components led us to create a beam that was strong enough to carry the load while remaining lightweight. Likewise, the middle bulkheads were optimized to have weight-reducing holes designed into the prints used. Finally, the thickness of the skin was optimized to be thick enough to successfully distribute the weight of the applied load, while not becoming a massive weight gain for the wing. Optimizing all of these different features made it possible to ensure that the final wing design was successful, bearing the load while remaining lightweight.

As part of optimizing the different components of the wing, it was necessary to determine the ideal materials that each part should be assembled from. The central beam used aluminum 7075 T6 as it is high-strength and does not deform greatly under load. However, for the outside bulkheads and the skin on both the top and bottom surfaces, it was decided to use aluminum 6061 T6. In the case of the bulkheads, the 6061 T6 was lighter and more easily deformed, which enabled the rivets to expand into the material successfully. In the case of the skin, using the 6061 T6 would allow for the skin to deform with the application of the load. This deformation would ensure that the load was distributed evenly across the skin. Finally, the group used rivets of various depths to attach the various components. It was found that to maximize the rivets, adding as many rivets as possible to the first bulkhead would minimize the stress on each rivet. After ordering all of the parts for the wing, the total cost came out to \$267.30.

These materials were then assembled into the final wing. There were several small issues, such as needing to try twice to successfully manufacture the central beam and needing to manually readjust the first bulkhead before attaching it to the beam. However, these issues were easily remedied by making small alterations to the wing design during the manufacturing process. Despite these changes, the group believes that the wing was still nearly as strong as it was in the modeled version on CREO.

The final assembly of the wing weighed 1.88 pounds, and when tested, it successfully held the required 95.5-pound load. As the loads increased, it was clear that the skin surfaces were buckling. After more buckling, the wing failed at 180 pounds, as the beam broke, the leading edge flew off, and many rivets popped off the wing. The calculations that were run in CREO were approximately correct, as the wing failed at around 1.88x the design load, and our estimations on the failure of the wing were purposefully kept lower to ensure that a safety factor was present in the design. However, it may have been possible to reduce the weight of the wing while still surpassing the necessary 95.5-pound benchmark.

If the group had further time and iterations to improve the design, the group would have liked to experiment further with reducing the thickness of the beam, reducing the weight substantially. Some examples of this might include thinning out the flanges or having the flange thickness scale down along the length of the beam. To ensure the wing would still survive, more rivets would have been placed into the beam along the first bulkhead, as well as more rivets being placed on the bottom skin surface, to prevent the buckling that occurred, and to better distribute the load onto the skin.

Another way we would try and reduce the weight would be to reduce the number of bulkheads or reduce the size of them. This is because we had the bulkheads evenly spaced and designed them to fail at approximately 100 pounds. But despite these efforts, the plastic did not deform within the wing until a 155-pound load was applied. So, reducing the number of bulkheads may help to reduce the weight of the wing. In addition to this, it might be possible to further remove weight from the wings by removing more material from the bulkheads themselves.

Lastly, one place where we could have decreased the weight was on the bottom side of the wing. We debated between choosing a skin thickness of .04" or .03" of AL 6061. While we did not rivet the back side completely, it was not possible to see if the bottom thickness was overkill or not. In a future iteration of this design, we would have liked to see what would happen if the thickness was .03" on the bottom with those rivets fully attached. Other variations to the skin in future iterations might further alter the thickness of the skin or change the material that the skin is constructed from.

These small alterations could have a significant effect on the performance of the wing, and should be examined more closely in the future. Despite the potential for further weight reduction of the wing, the group is ecstatic about the performance. In particular, the group is proud that the load was distributed along the skin and the components failed in the manner and locations that we expected them to. The group was able to work well together as a team, distributing analysis work, manufacturing, testing, and other key tasks among the group. The collective action of the team allowed us to find success in this project. Beyond this, the group gained an appreciation and understanding of the importance of computer analysis, as well as clear communication and collaboration. These skills and experiences will be invaluable as the group moves further along in undergraduate education and into careers beyond Princeton.

Individual Contributions

Every member of the AUTOBOTS team was crucial to the success of the wing that was manufactured and tested. Throughout the whole process, everyone contributed. This section will go individual by individual and detail their contributions to the project.

Lucas - Team Lead

I led the overall vision and decision-making process for the wing along with Jack. We helped guide the project from any initial designs to the final tested piece. During the optimization phase, I helped with the modeling of the rivets for analysis. These were really difficult to model directly, and helped to identify an approach that captured their structural behavior. I also assisted in different truss designs for the beam and helped coordinate between the beam and bulkhead subteams to communicate with some design decisions, including the cross-sectional area of the beam and the placement of the first and last bulkheads. Additionally, I helped attempt to make a skeleton model for the bulkheads, because at first it was difficult to get them to regenerate

properly. I helped guide the decision-making process for the skin material and thickness, where I helped conduct stress analyses to make informed designs about the skin. Facilitated the writing of the PDR, which everyone helped tremendously with. I assisted with machining the first iteration of the central beam and attempted to fix any dimensional and tooling errors. Later, I helped prepare the final skin tracing for cutting, including repairing issues from an initial failed cut. I also took part in riveting the bulkheads during the final assembly. Lastly, I helped coordinate and write the PDR, which Jack wrote/edited as well, and we got input from the group about their individual sections.

Jack Byrnes - Team Lead

Oversaw the analysis of the wing, ensured there was parity and close communication between the various subteams, and performed quality control on the designs to ensure the wing design would be successful. This quality control included assessing the practicality of manufacturing the various parts, as well as conducting analyses within the CREO software to ensure that the final design would still be capable of surviving the necessary load. I was responsible for preparing the central beam for CNC milling. Despite running into challenges the first time the beam was prepared, I was then able to make a successfully prepared beam on the second attempt. I also helped Glenn when running the CNC mill, helping to find the beam edge, and operating the mill itself. I then worked with Raj and Glenn to use the waterjet and cut out the external bulkheads. In this process, I learned how to actually use the waterjet equipment, and how to transform a CREO part into something that the waterjet can recognize. I then led the final assembly of the wing, overseeing and helping lay out and cut the skin, fit on the bulkheads and leading edge, and ensure that the beam was still up to spec after manufacturing. Finally, I oversaw and helped to complete the installation of rivets into the wing in order to hold it in one piece. To this end, I was responsible for determining the rivet depth, ensuring that the drill did not destroy any of the parts in the wing, and installing the rivets themselves. Finally, I played a large role in the creation of the PDR and FDR, alongside writing the majority of the introduction and conclusion, I also read over and edited each category to ensure that the report accurately explained the group's development process.

Tyler Harris - Beam Lead

I, along with Sawooly, was in charge of the beam subteam. I was tasked with designing the beam for the wing, and in order to do this, I used the beam from the original wing assembly that we designed together in class as a base model. When designing the beam, my goal was to optimize it in a way that would result in the best combination of strength and simultaneously reduce its weight. To ensure strength, I assigned aluminum 7075 instead of aluminum 6061 as the beam's material and kept the width of the web relatively large and constant along the beam to prevent the beam from failing under torsion. To reduce weight, I created circular webbing along the web of the beam as a truss design and maximized the size the holes could be while still ensuring that the beam would not fail. In order to confirm my optimization, I used the simulation

software in Creo Parametric to run analysis tests on the beam with a force load of 100 lbs on the end. This allowed me to check if changes to certain dimensions would cause the beam to fail, as I aimed to increase the distribution of the stress along the beam while keeping the stress on all parts of the beam below 60 ksi. I repeated this process of using the simulation tool until the beam was optimized. I then created the Creo file and code for the beam to be properly made in the CNC machine. I also helped write the beam sections of the PDR and FDR, including the pictures of the beam's force load analysis and engineering drawing. I helped to assemble the beam with the bulkheads and leading edge.

Rajan Patel - Bulkheads Lead

My main contribution to the design of the wing was leading the Bulkheads team. First, I was able to design and simulate various combinations of bulkhead profiles for the front and end bulkheads which were used to determine the skin area of the wing. The second focus was on the spacing and number of bulkheads to maximize the distribution of stress along the wing and also help the rivet team decide where and how many rivets to use. Throughout all of my analysis I focused on keeping the stress on the front and end bulkheads below 45 ksi which were made out of aluminum 6061. Utilized a skeleton model to enable greater efficiency in iterative designs of the bulkheads as well as created drawings for each part. I also worked on optimizing the cutouts from the middle bulkheads as well as choosing the materials which ended up being PLA for the 3d printing and the 6061 front and end bulkheads to reduce weight while maintaining structural integrity. In the design process I also created the leading edge which was also 3d printed using PLA. I also worked extensively on the manufacturing of the wing from 3d printing all the bulkheads and leading edge as well as working alongside Jack to water jet the front and end bulkheads. Helped rivet the wing together as well as cut the top skin of the wing. I also worked extensively on the PDR and FDR helping write numerous sections and proofreading. Overall my contributions were both analytical with CREO simulating and designing the wing to meet the specific requirements as well as manufacture in which I created and ensured the parts met specifications.

Esteban Nunez Perez - Rivets

I was primarily responsible for design decisions relating to the riveting of our wing. This meant extensive amounts of time were spent creating an adjacent wing model that could be utilized in Creo to analyze how rivets would react under stress, and actually conducting the analysis using Creo simulation features – as seen in the Rivets portion of the appendix. Also, I wrote the rivet portion of the PDR, much of which was explaining the model I made to analyze the rivets and assumptions that had to be made in regards to said model. When it ultimately came down to the manufacturing many decisions regarding the riveting were not done based on the analysis conducted in Creo, this was due to a variety of reasons, but most prominently due to the fact that within Creo it is difficult to accurately model how rivets function in a complete assembly. So when deciding on locations for the rivets, I decided that 4 and 4 on the top of the first and second

bulk heads would be sufficient. I decided that the middle bulkheads on the bottom face did not require a ton of riveting, so only having at least one down the middle seemed appropriate. In retrospect, the group and I decided that having those rivets to close the bottom skin to the leading edge would have helped with the buckling and deflection. I personally worked on manufacturing a small portion of the rivets but that job was mostly completed by our team leads due to other time commitments on my part.

Ani Kozak - Skin

I, along with Connor, was responsible for the skin subteam of the wing. Our task centered around evaluating and optimizing both the material and thickness of the wing's external surfaces to meet the expected design constraints on deflection and stress. We began by testing various combinations of AL6061 and AL7075 aluminum alloys at a 0.02-0.04" range thickness, but simulations showed that the deflection consistently exceeded our 1-inch threshold. After analyzing stress and deflection across multiple configurations, including mixed material top and bottom skins, we determined that a uniform 0.04" thickness of AL6061 provided the best balance of ductility, strength, and weight. I contributed to running and interpreting these simulations to help finalize our material decision, which I outlined in the skin section of the PDR. During the manufacturing process of the wing, I helped trace and cut the wing shape twice from the aluminum sheet using the band saw and ensured bulkhead alignment before marking the cut lines. To secure the trailing edge, we taped the skin together, creating a flush and continuous surface that improved structural integrity. During the final load testing, we observed localized bending near the leading edge due to missing rivets, which we noted as a key area for improvement. Overall, I played a central role in both the simulation-driven material selection process and hands-on construction of the skin, helping ensure the final wing met our desired performance expectations.

Connor Brown - Skin

Working along with Ani, we were both tasked with finding both the material, and thickness for the top and bottom skins that would reduce weight without sacrificing the strength of our wing. At first we began to experiment with a mixture between AL7075 and AL6061, trying each side as one or the other and then on to both. Among all these different combinations of skin material, we also tried every one of these analyses with different thicknesses. At the end of our analyses we found the best ratio for weight to deflection was to use AL6061 on the top and bottom with a thickness of 0.04". Personally I ran about half of the combinations in analyses and Ani ran the other half, working together with her to figure out which combination would be best. For the PDR both Ani and myself described what our thought process was in detail and how we came to the conclusion we did. With regards to the actual manufacturing of the wing, I mainly used the band saw to cut out the skin so that it could fit our beam and bulkheads with no problem, this was made easy as Ani had already traced out the shape for me. The majority of my

job on this team definitely was done in CREO working on analyses and changing our design, but some of it was in the actual manufacturing.

Sawwooly Li - Beam

In this project, I aided Tyler in the process of designing the beam, utilising the I-beam which was utilised in class as a base. In this procedure, I explored various design modifications, including adjustments to the flange thicknesses, overall shape, and proportions, with the goal of maximizing the structural strength of the wing whilst reducing its weight. These decisions were assessed through analysing data produced by CREO Simulate, and I spent a lot of time experimenting with different patterns along the length of the beam, including triangular and vertical supports, in which we ultimately chose the circular webbing due to its superior weight and strength tradeoff. In the shop, I also contributed to the manufacturing of the wing by 3-D printing Raj's bulkheads with PLA on the shop's printers, and furthermore aided in the process of creating the skin, aligning the bulkheads and riveting them together. I measured and drew the dimensions on the Aluminium sheet for the skin, and personally cut the bottom skin of the wing to the required specifications. Beyond the physical construction, I also helped to write the PDR and proofread the document to ensure clarity, coherence, and technical accuracy. I also made certain that all figures were correctly referenced and thoroughly analyzed in the report to support our design decisions.

Appendix A: Images and Analyses

Beam

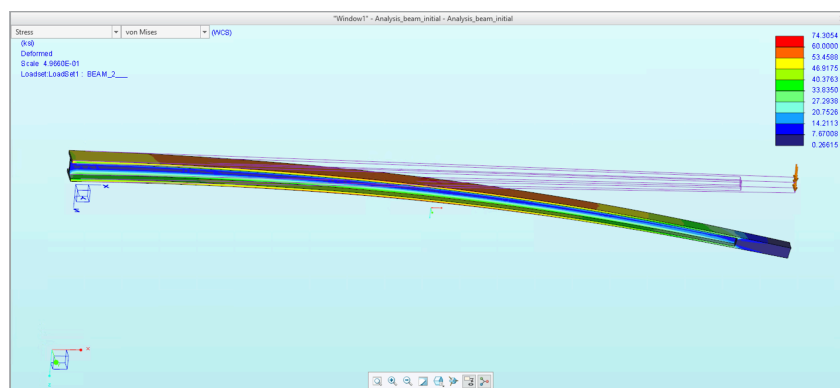


Figure 1A: Initial Beam Without any Weight Reduction

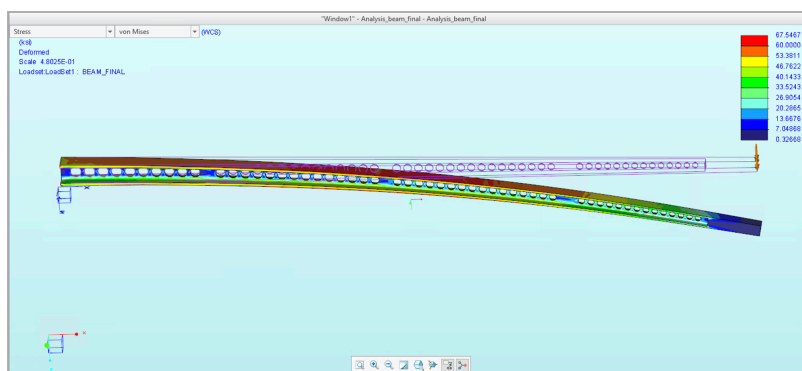


Figure 1B: Final Beam, with Weight Reduction

Rivets

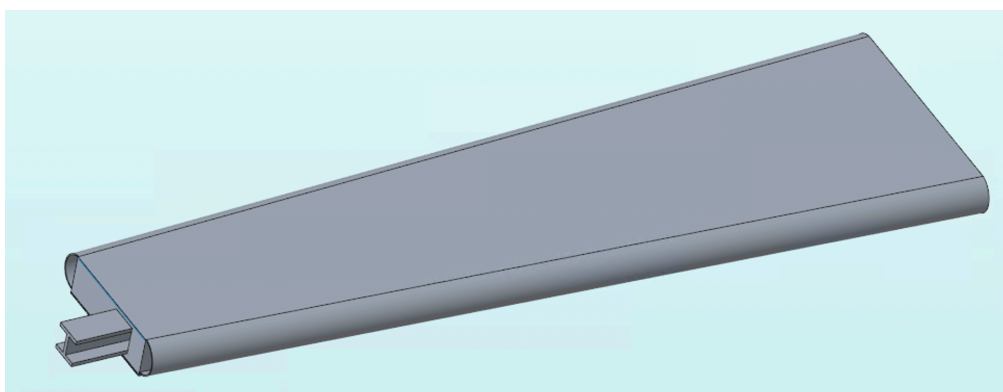


Figure 2A - Wing model used for rivet analysis

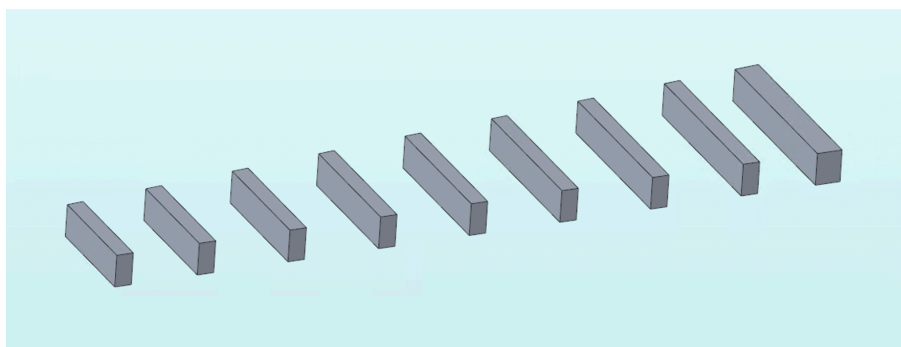


Figure 2B - Bulkhead part used for rivet analysis

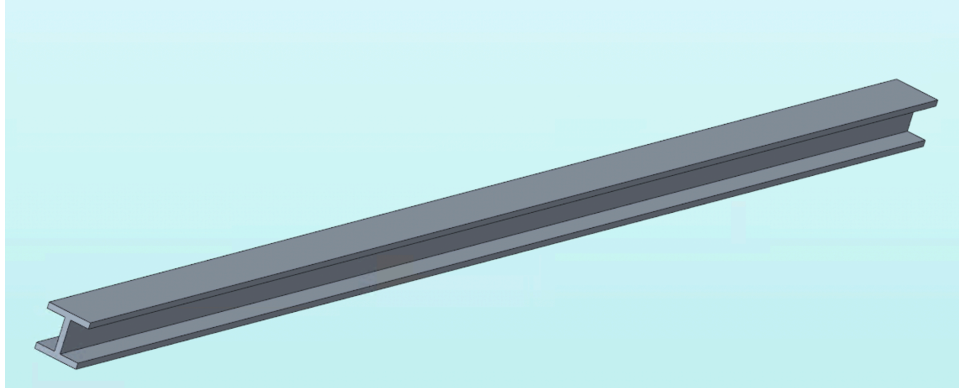


Figure 2C - Beam part used for rivet analysis

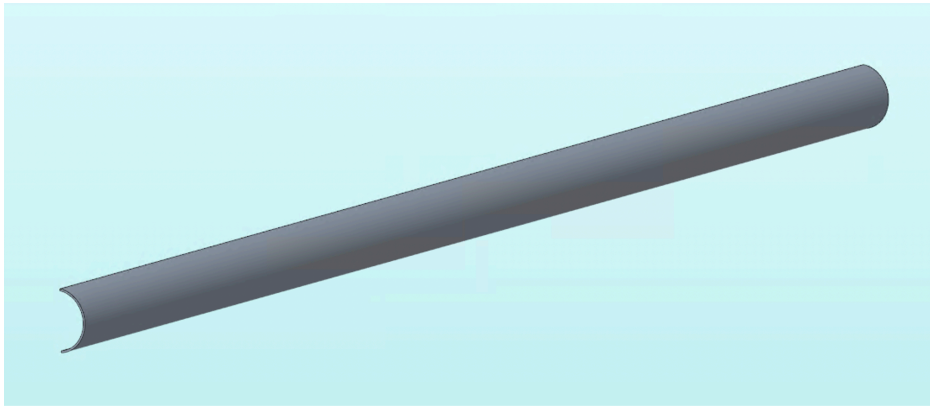


Figure 2D - Curve part used for rivet analysis

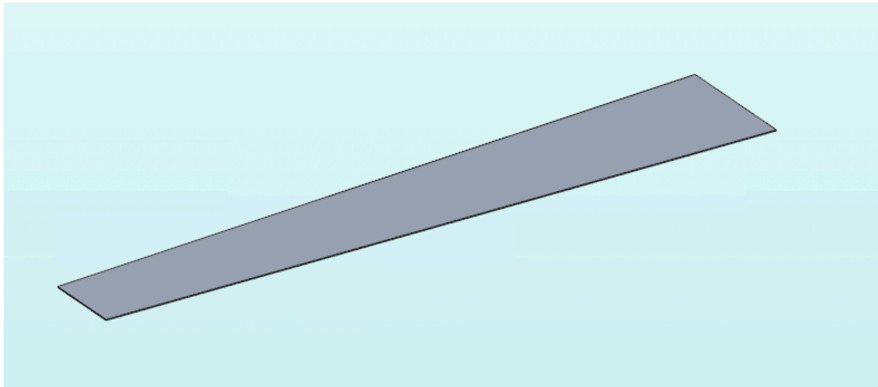


Figure 2E - Wing surface part used for rivet analysis

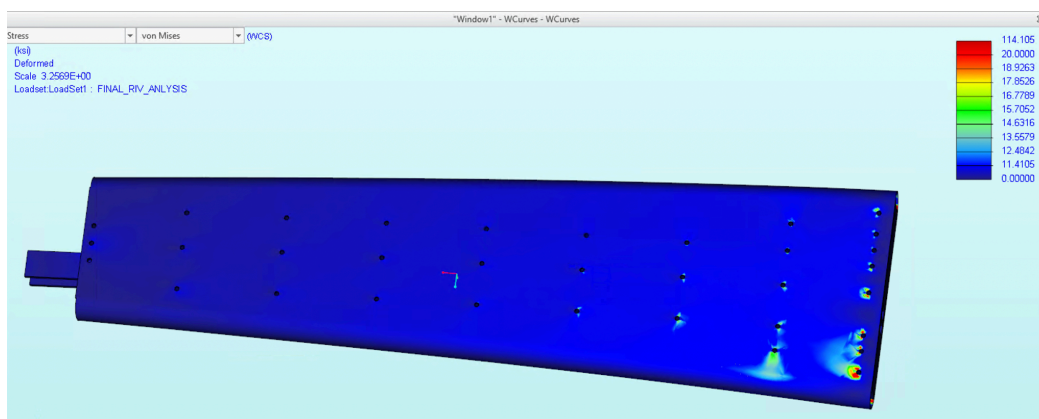


Figure 2F: Final rivet analysis

Skin

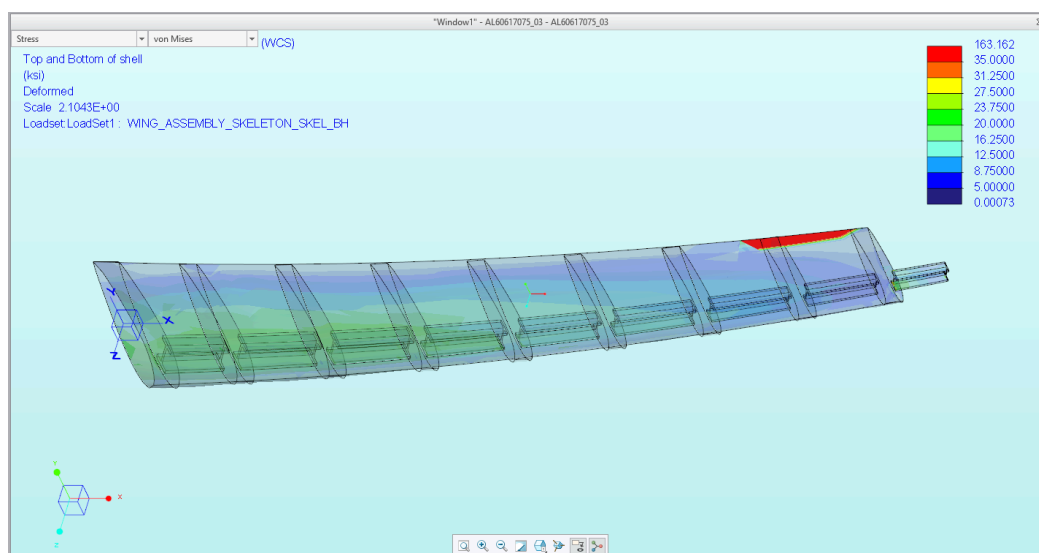


Figure 3A: Unoptimized Skin Stress Distribution (AL6061 top, AL7075 bottom, .03")

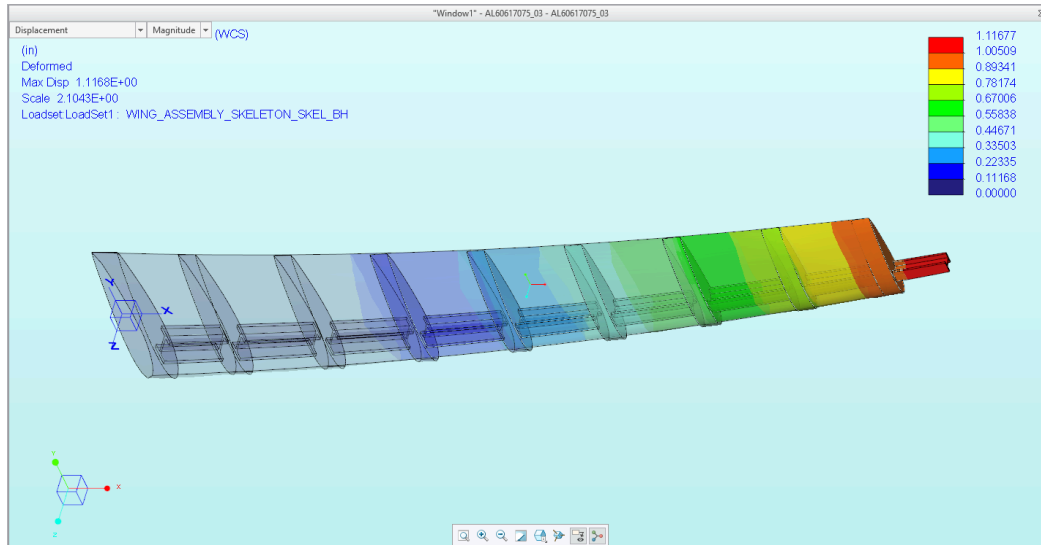


Figure 3B: Unoptimized Skin Deflection Analysis (AL6061, .03")

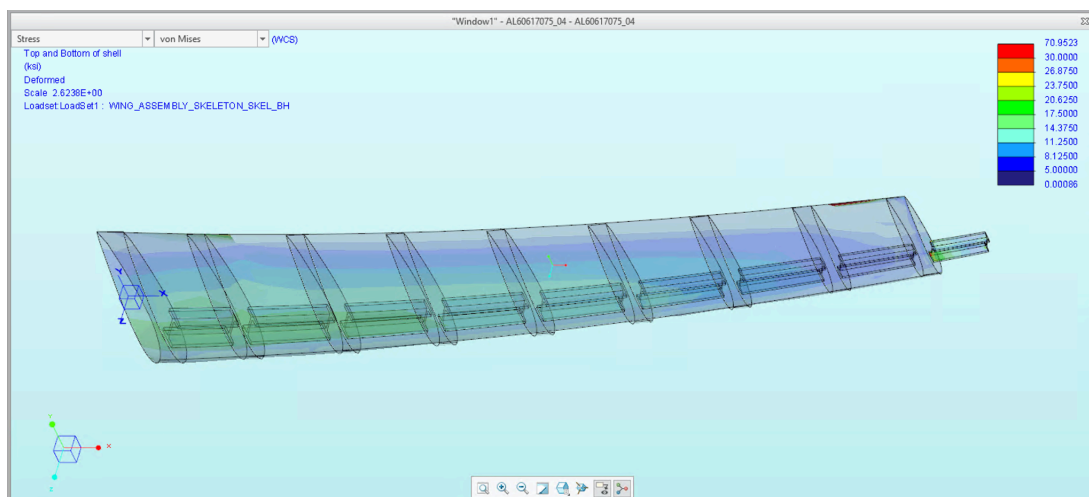


Figure 3C: Heavy Skin Stress Distribution (AL6061 Top, AL7075 Bottom .04")

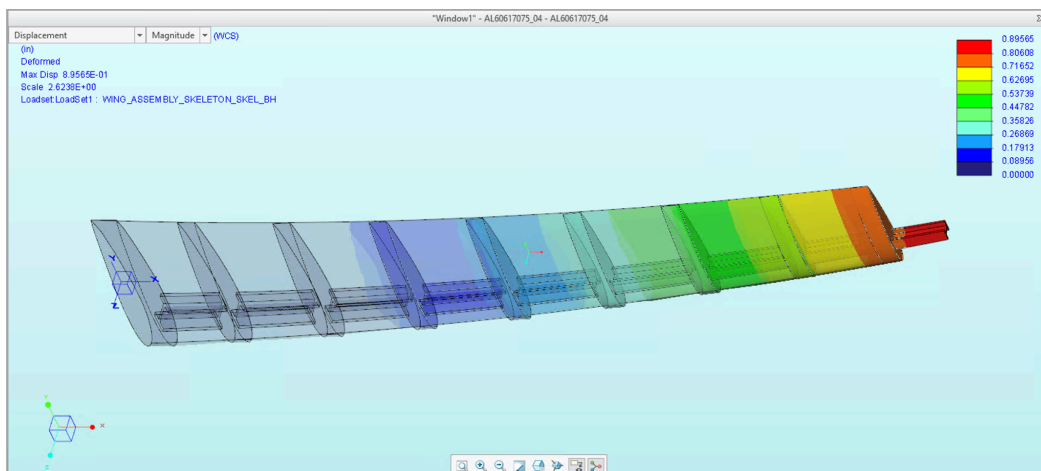


Figure 3D: Heavy Skin Deflection (AL6061 Top, AL7075 Bottom .04")

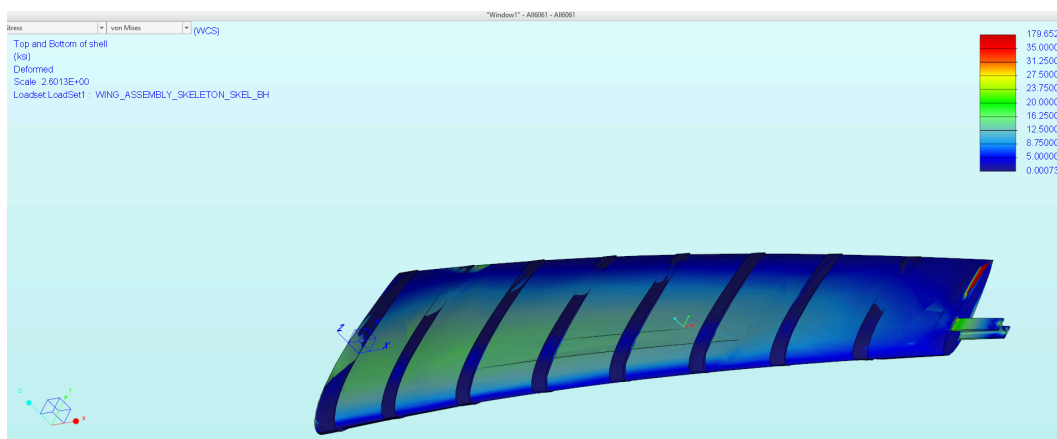


Figure 3E: Finalized Skin Stress Analysis (.04", AL6061)

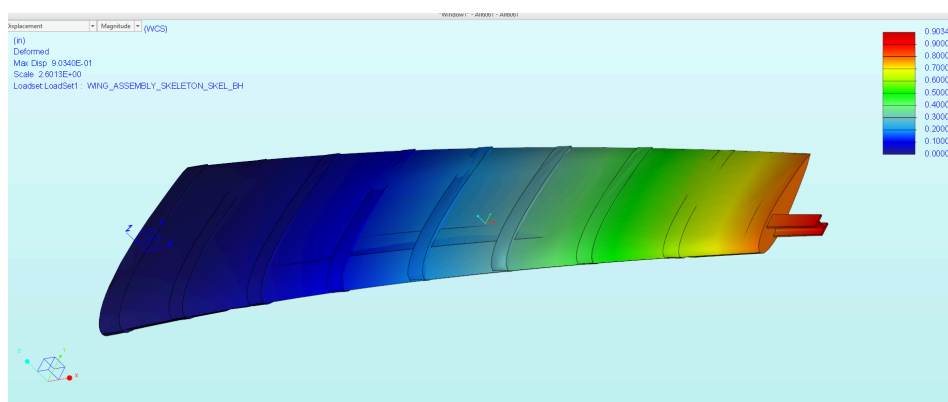


Figure 3F: Finalized Skin Deflection Analysis (.04", AL6061)

Bulk Heads

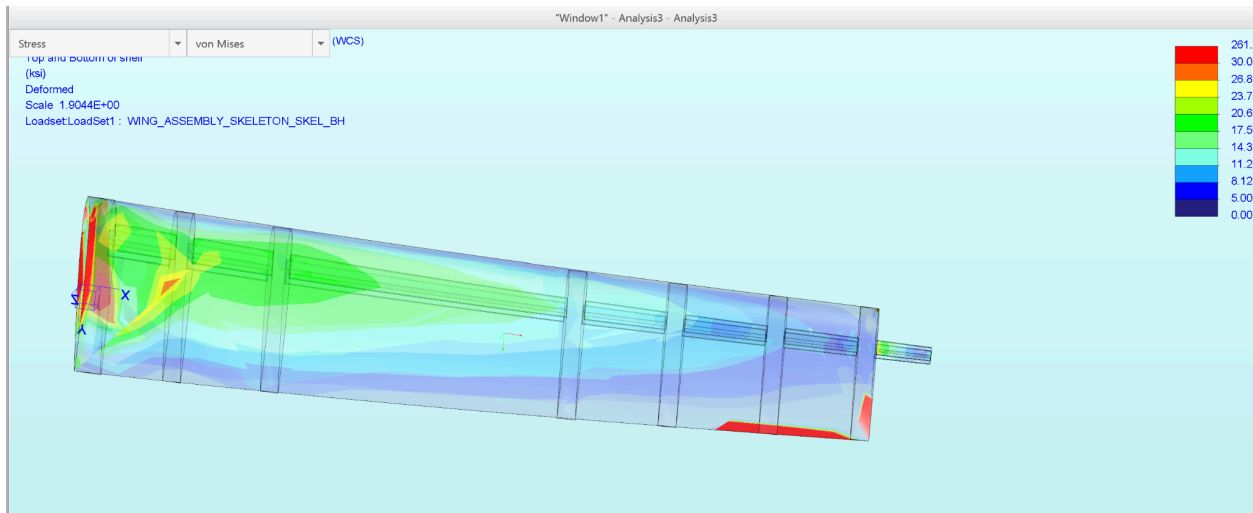


Figure 4A: Initial Bulkhead Design

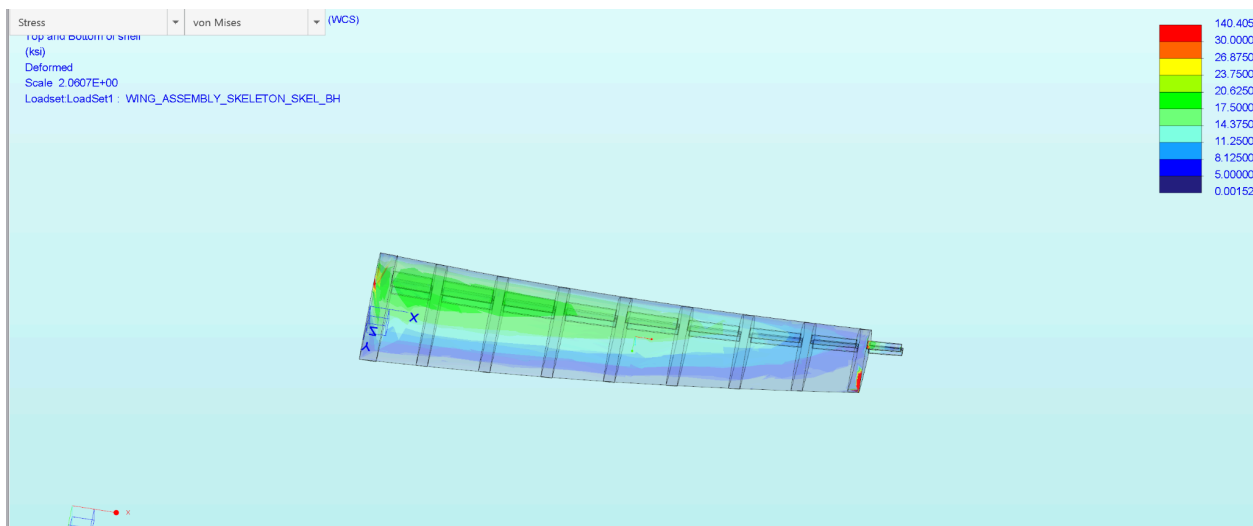


Figure 4B: The Final Bulkhead Spacing

Front Airfoil Scaling (BigNum) Keep 2 to 1 ratio in section 1 of bulkhead sweep blend	Rear Airfoil Scaling (Big Num) Keep ratio of 2-1	Wing Area	Maximum Stress @100 lbs	Deflection @100 lbs
6	3	102.832	~210 ksi	~1 in
5.5	3	97.1136	237 ksi	1.14994 in

5.5	3.4	101.68	212.303	1.063990
5.2	3.5	99.3914	136.764	1.14414
5	3.8	100.5	139.294	1.164
5.8	3.1	101.68	256.8	.97339

Table 4.1: Table Detailing Specific parameter for the Bulkhead Optimization

Further Analysis

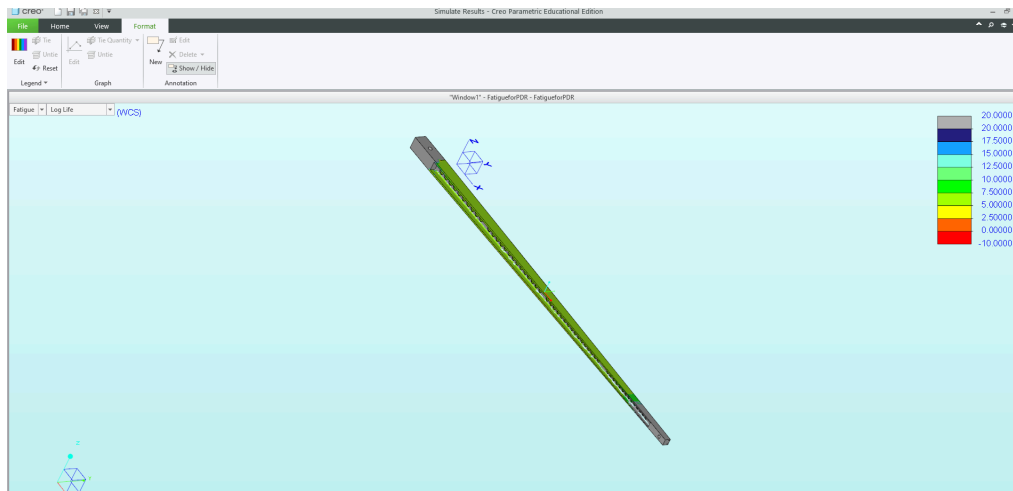


Figure 5A: Beam Fatigue Analysis (Top Side)

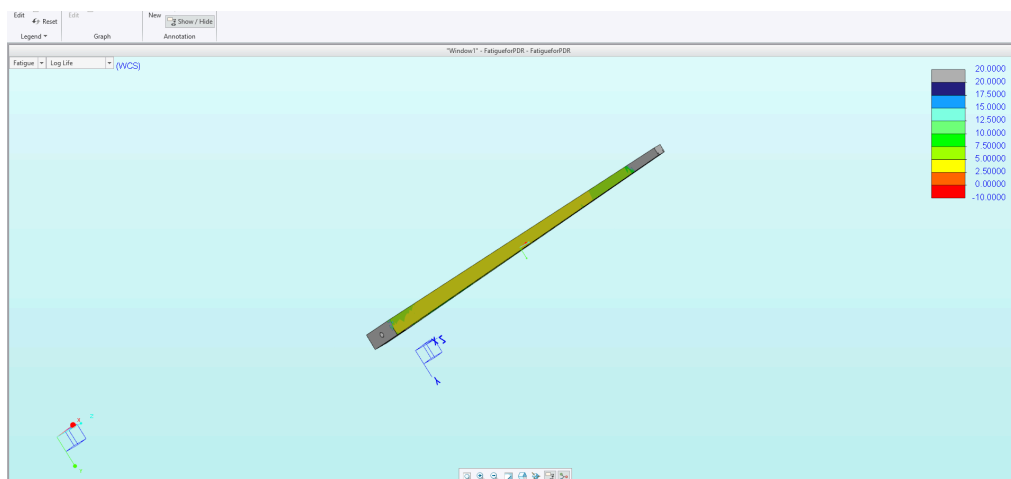


Figure 5B: Beam Fatigue Analysis (Bottom Side)

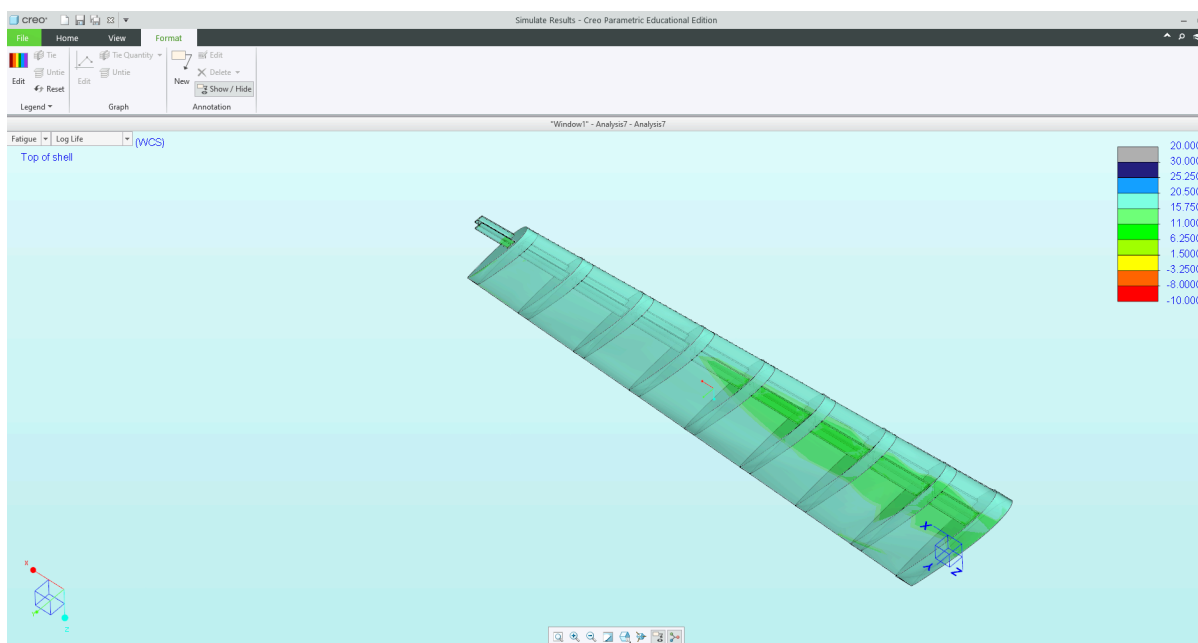


Figure 5C: Wing Fatigue Analysis

Testing



Figure 6A: Final Wing Design

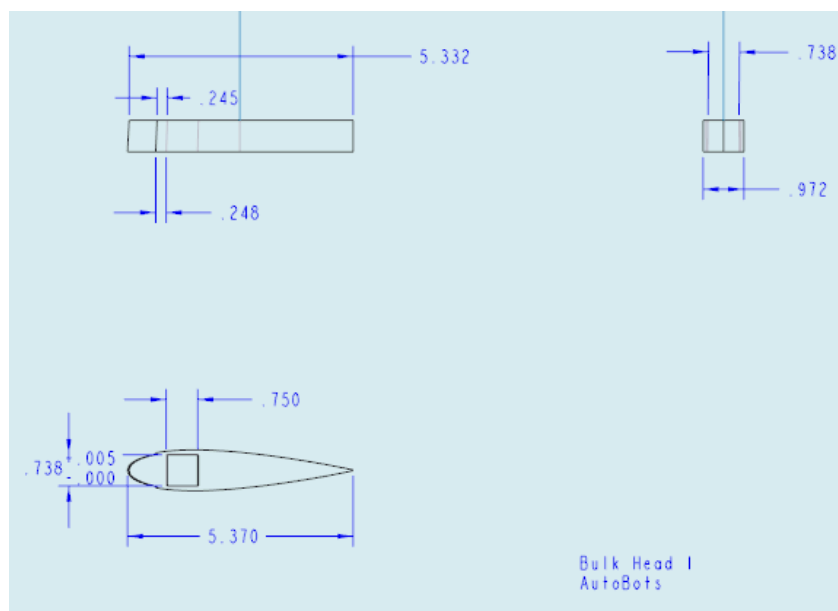


Figure 6B: Aftermath of 180 lb Test

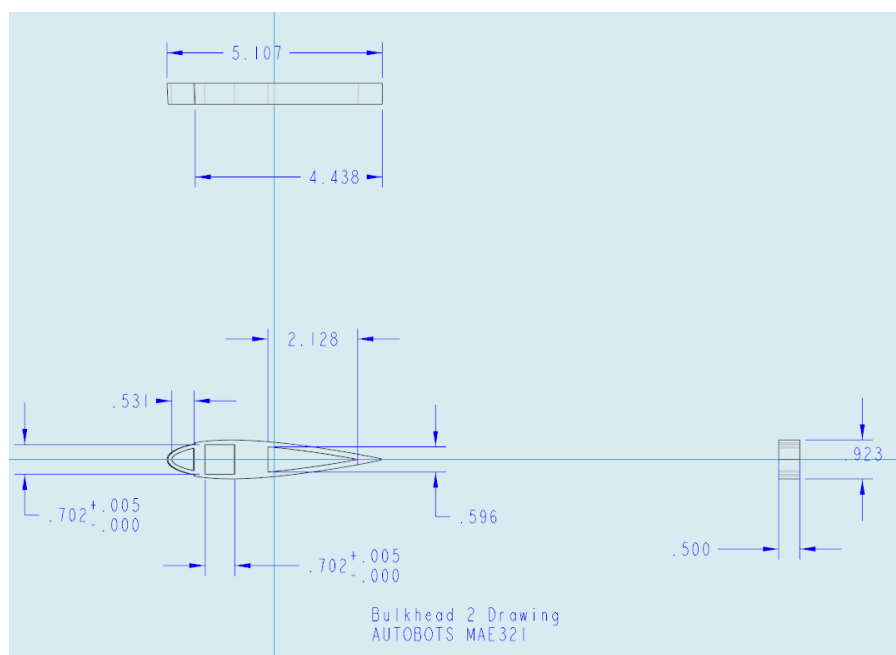


Figure 6C: Look into the beam failure

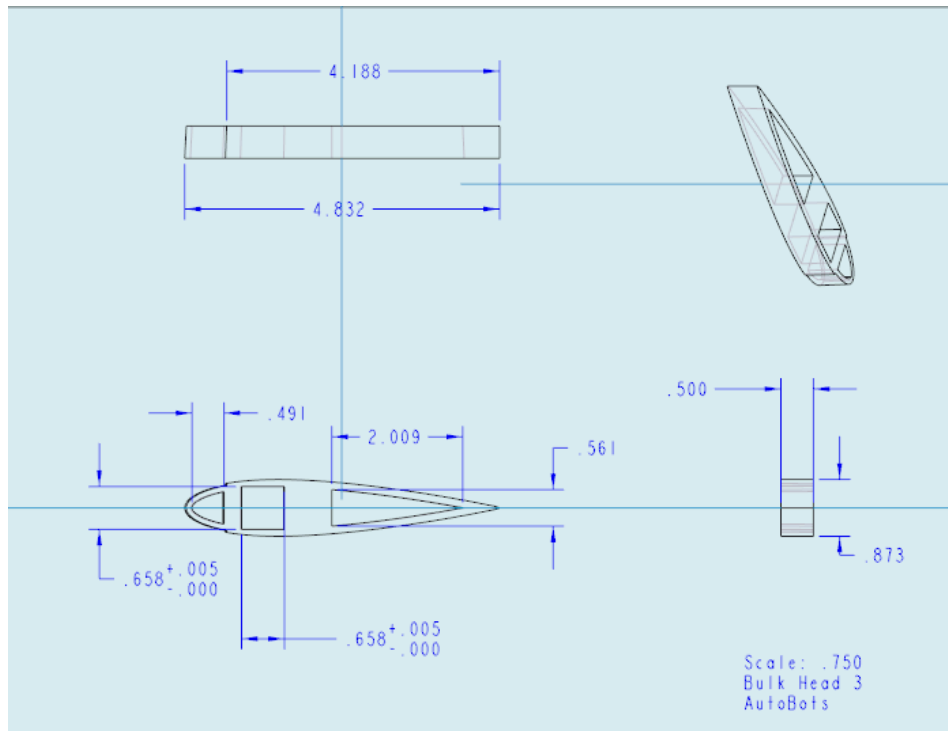
Appendix B: Engineering Drawings



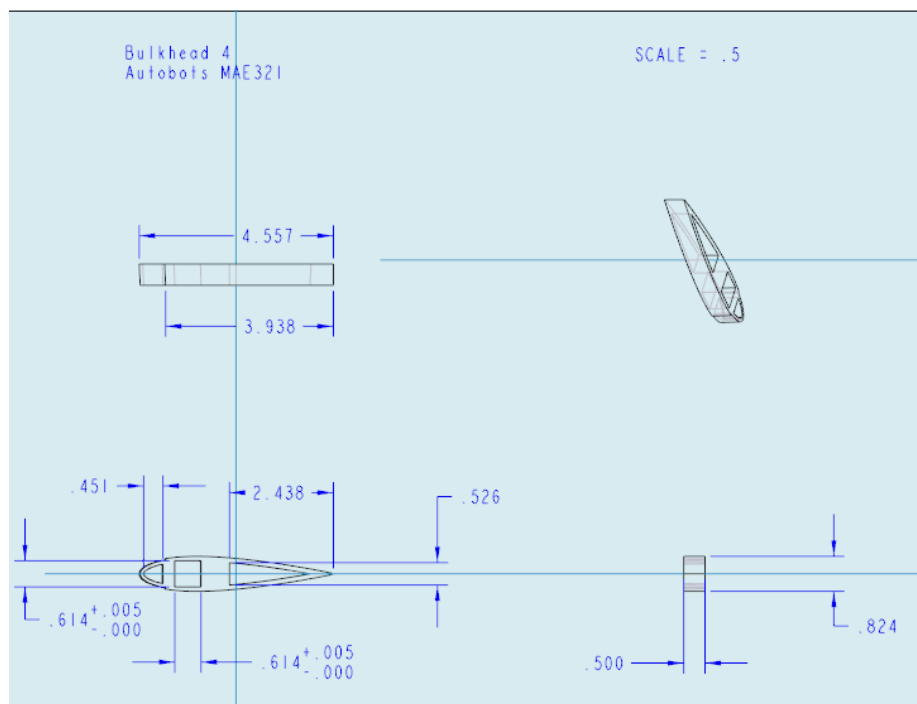
Drawing 1: Bulkhead 1



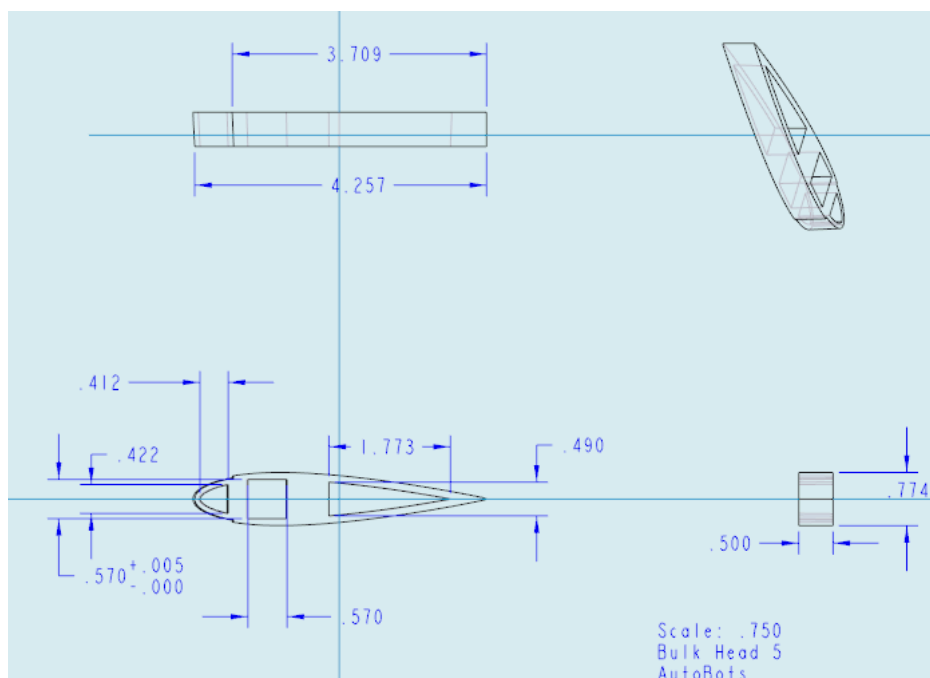
Drawing 2: Bulkhead 2



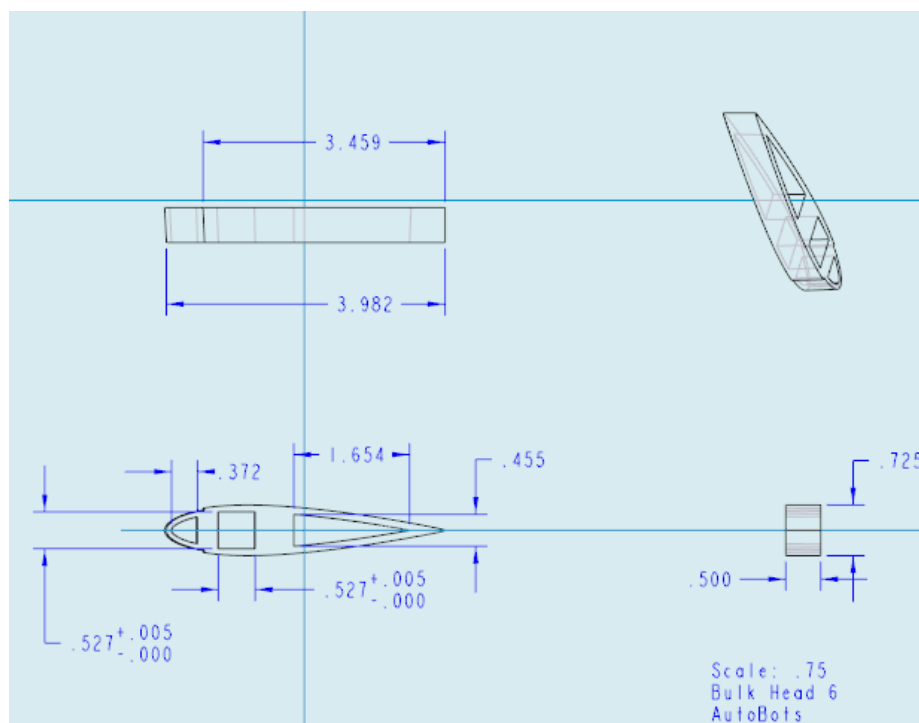
Drawing 3: Bulkhead 3



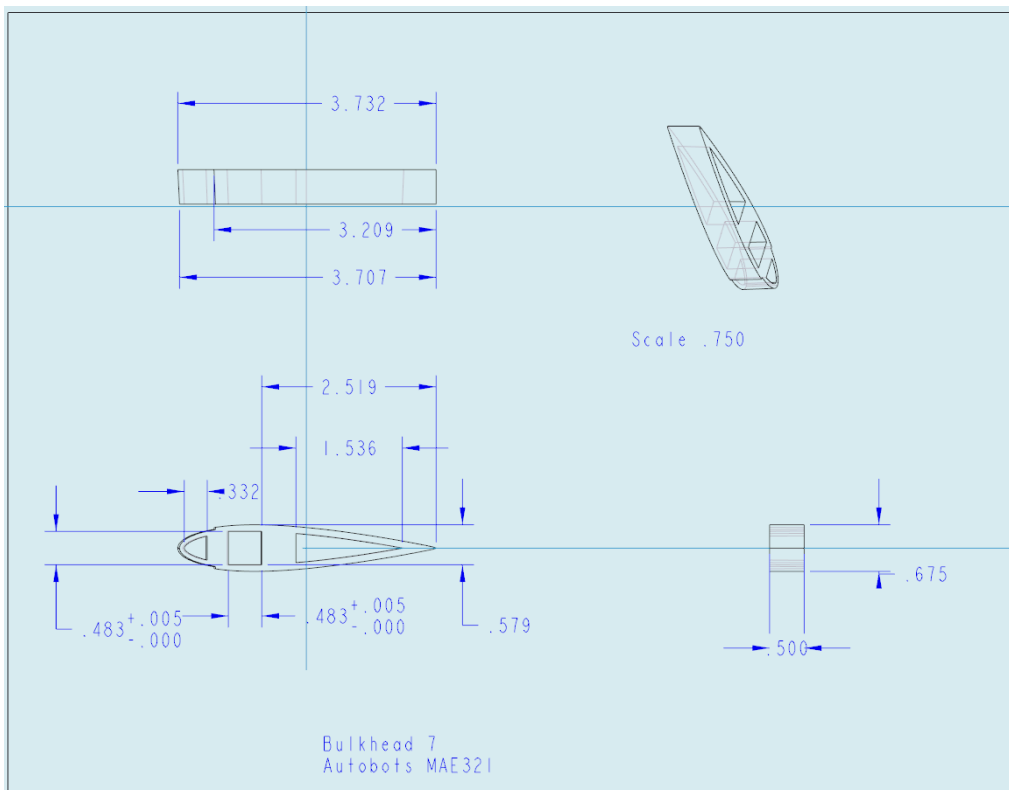
Drawing 4: Bulkhead 4



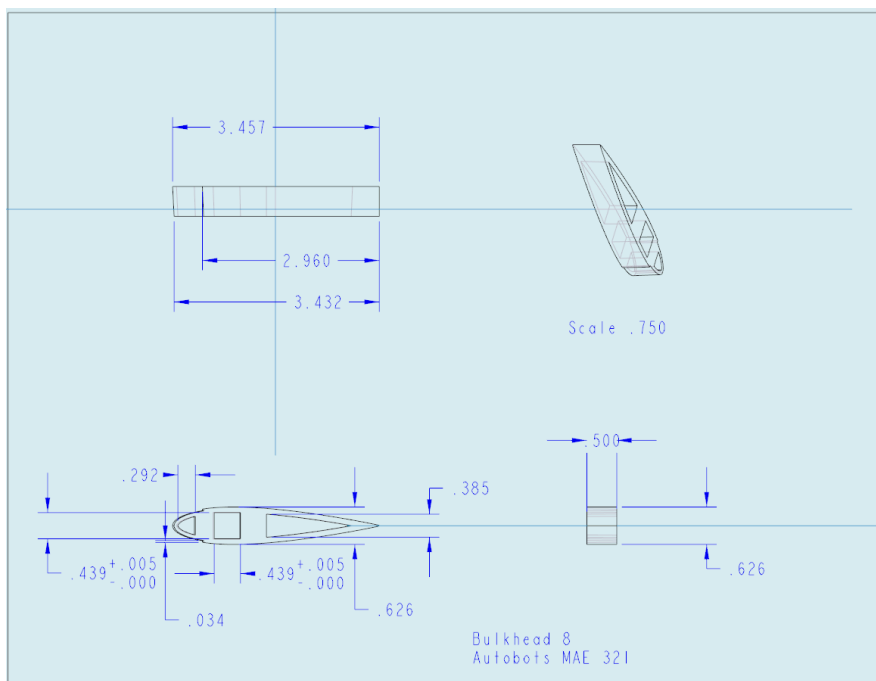
Drawing 5: Bulkhead 5



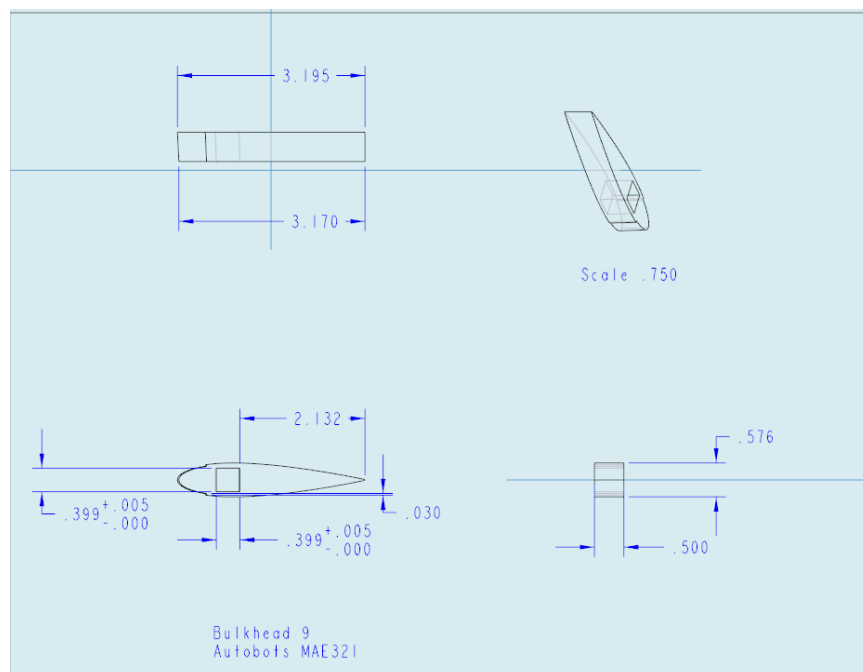
Drawing 6: Bulkhead 6



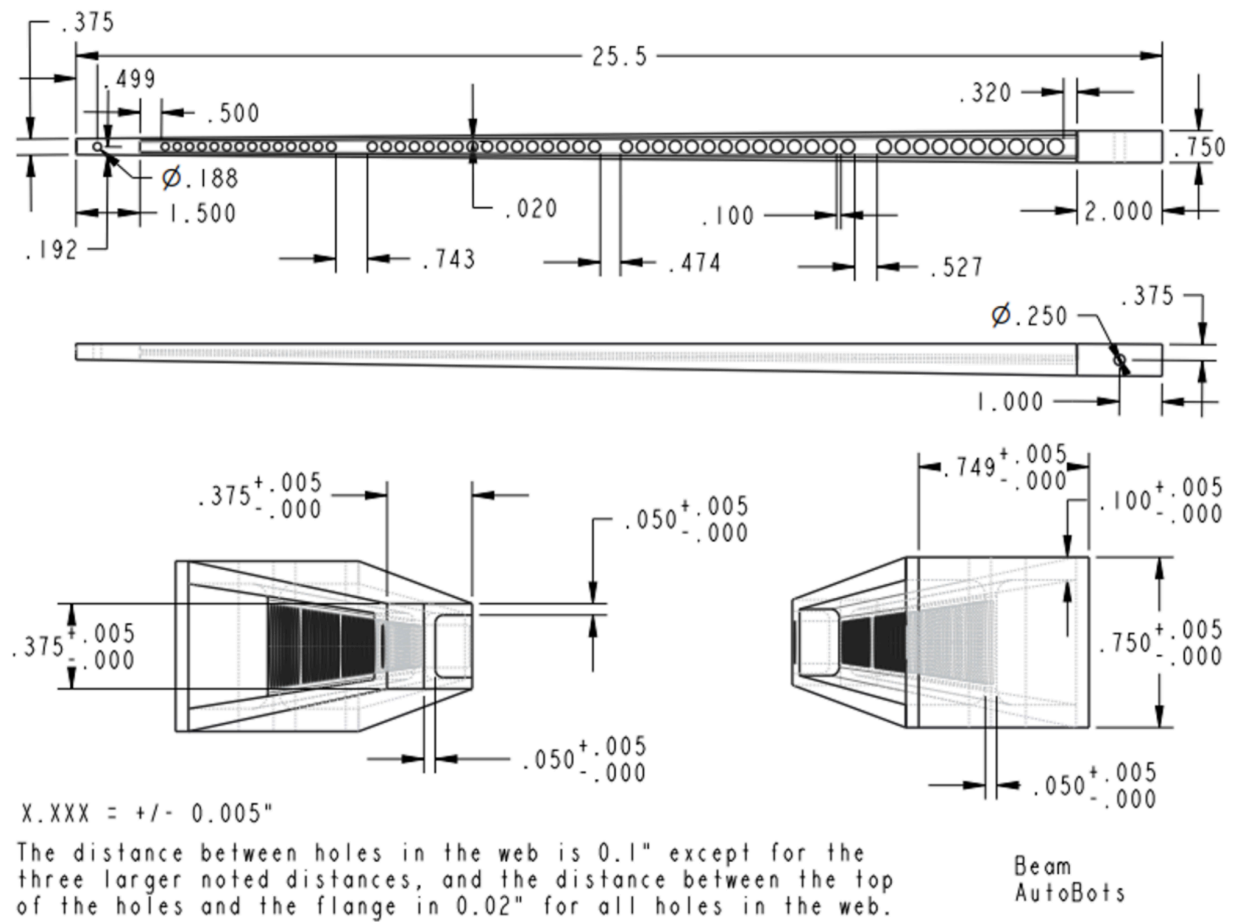
Drawing 7: Bulkhead 7



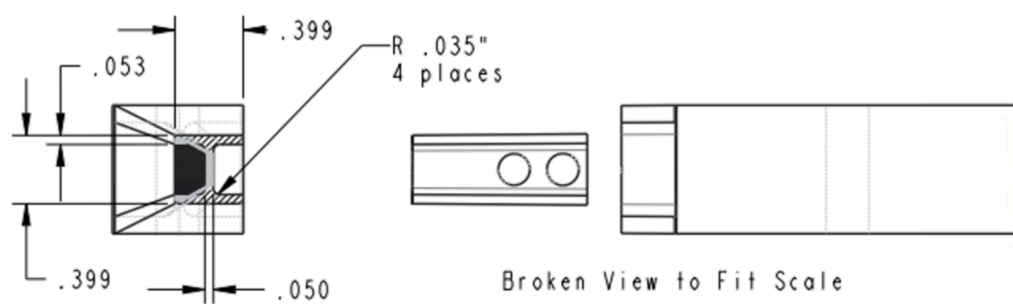
Drawing 8: Bulkhead 8



Drawing 9: Bulkhead 9



Drawing 10A: I Beam Final Drawing



X.XXX = +/- 0.005"

Beam
AutoBots

Drawing 10B: I Beam Final Drawing